MERN Stack Machine Test

Brand Management CRUD Application

Task Overview

Build a **Brand Management Application** using the **MERN stack**, where a registered user can log in and manage brand records including **brand name and image**.

This task is designed to evaluate the candidate's understanding of **MERN concepts**, clean project structure, API design, and basic frontend–backend integration.

Evaluation Focus Areas

- REST API design (Express + Node.js)
- MongoDB schema design using Mongoose
- Registration & Login
- CRUD operations
- File/Image upload handling
- React component structure
- State management
- API integration using Axios / Fetch
- Clean code & folder structure

Functional Requirements

1. User Authentication

- User must be able to **register** with:
  - Username
  - Email
  - Password
- User must be able to **log in** using:
  - Email
  - Password
- After login, user should be redirected to the **Home/Dashboard** page.

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2. Brand Management (CRUD)

After login, the user should be able to manage brands.

Each **Brand** should contain:

- Brand Name (text)
- Brand Image (file upload)

The user should be able to:

- Create a new brand
- View a list of brands
- Update an existing brand
- Delete a brand

Brand list should be displayed in a **table or card-based UI**.

CRUD operation (Create / Update / Delete) must be handled **without page reload** (SPA behavior).

Technical Requirements

Backend

- Node.js with Express
- MongoDB with Mongoose
- Multer (or similar) for image upload
- Proper API routes & controllers
- Basic input validation

Frontend

- React
- Axios / Fetch for API calls
- React Router for navigation
- Basic form validation
- Simple, clean UI (Bootstrap / CSS / any framework)

Project Structure Expectation

- Separate **frontend** and **backend** folders
- Logical folder separation:
  - routes
  - controllers
  - models

Submission Instructions (IMPORTANT)

- The completed project must be **zipped** and submitted.
- The ZIP file should include:
  - Frontend folder
  - Backend folder
  - Any required configuration files
- **Do NOT include** node_modules inside the ZIP.
- The ZIP file must be sent via email to ejessyjose@gmail.com.

Failure to follow submission instructions may result in **disqualification**.

Duration

- **2 - 3 hours**

Important Notes (Be clear as interviewer)

- UI perfection is **NOT** the goal.
- Clean logic, correct flow, and working functionality **ARE** the goal.
- Copy-paste boilerplates without understanding will be obvious.